

Gwendal Beaumont

PhD student in Computer Sciences

Brest, France

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Education

- 2023 – **PhD — Computer Sciences, IMT Atlantique**, Brest, France
Title of the thesis: TwinDevOps, an automated process for the deployment and dynamic evolution of Digital Twins.
- 2020 – 2023 **Engineer's Degree — General Engineering, IMT Mines Alès**, Alès, France
One and a half year of general training and one and a half year in the Department of Software Engineering and Artificial Intelligence.
- 2018 – 2020 **Classe préparatoire aux grandes écoles, Lycée Malherbe**, Caen, France
Two-years intensive training to prepare for the French engineering schools competitive exam. Fields of study: Mathematics, Physics, Computer Sciences.

Experience

- 2026-01 – **Doctoral Visiting Researcher, Technische Universität Wien**, Vienna, Austria
2026-04 Doctoral mobility hosted by Assoc. Prof. Bork Dominik where I worked on an extension of a process developed during my PhD thesis.
- 2023-03 – **Intern, ALTEN**, Rennes, France
2023-09 Research & Development internship on the labelling process of data in order to be used with Machine Learning algorithms. Detailed achievements:
 - Creation of a framework using Java Spring Boot to manage unlabelled data and get statistics from it
 - Creation of an interface using TypeScript and the Electron framework
- 2022-05 – **Visiting International Scholar, Miami University**, Oxford, OH, United States of America
2022-08 Internship as an assistant in the preparation of a master thesis on the effect of errors in agent-based simulation systems. Detailed achievements:
 - Creation of a simulation framework in Python to take errors into account
 - Creation of multiple agent-based simulations in Python to test our framework

Services

- 2026 **Seminar co-organisation, IMT Atlantique**, Brest, France
I co-organised one day of the 2026 Annual Seminar of the Department of Computer Sciences. Planning of the seminar is available [here](#).
- 2024 – 2025 **PhD students representative at the Student Board, EULiST Alliance**
Institut Mines-Télécom PhD students representative at the european level in the EULiST Alliance. This alliance integrates over 200,000 students and 20,000 staff at 10 universities in 10 countries.
- 2023 – 2025 **PhD students representative in a workgroup, Institut Mines-Télécom**, France
IMT PhD students representative at the national level in a work group focused on respect and equality.
- 2023 – 2025 **PhD students representative at the board of directors, Institut Mines-Télécom**, France
IMT PhD students representative at the national level for two years. I took part at the board of directors and worked with the scientific director of Institut Mines-Télécom on ways to improve PhD environment.
- 2021 – 2022 **Communication Manager, IMT Mines Alès student council**, Alès, France
Member of the communication managers team for the student council during a year. I administrated the student council's website, managed social networks, and answered and moderated emails and comments on social networks.

Languages

French Native

Spanish Intermediate

Korean Basics

English Fluent (*TOEIC: 975/990*)

German Basics

Book Chapters

- [B1] Gwendal Beaumont et al. “DevOps for and by Digital Twins Leveraging Virtual Replicas in Continuous Software Engineering”. In: *Digital Twins*. Ed. by Bedir Tekinerdogan and Cor Verdouw. Elsevier B.V., June 2026, pp. 227–242. ISBN: 978-0-443-45573-5. DOI: 10.1016/B978-0-443-45573-5.00018-5.

Journals

- [J2] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. “A Framework for the Comparison of Errors in Agent-Based Models Using Machine Learning”. In: *Journal of Computational Science* 72 (Sept. 2023), p. 102119. ISSN: 1877-7503. DOI: 10.1016/j.jocs.2023.102119.
- [J3] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. “A Scoping Review of Three Dimensions for Long-Term COVID-19 Vaccination Models: Hybrid Immunity, Individual Drivers of Vaccinal Choice, and Human Errors”. In: *Vaccines* 10.10 (Oct. 2022), p. 1716. ISSN: 2076-393X. DOI: 10.3390/vaccines10101716.

Conferences

- [C4] Gwendal Beaumont et al. “Engineering Digital Process Twins: Towards a Methodology Based on MDE and DevOps”. In: *ACM/IEEE 29th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion 2026. Málaga, Spain: Association for Computing Machinery, Oct. 2026. ISBN: 979-8-4007-2903-4. DOI: 10.1145/3837062.3839100.
- [C5] Gwendal Beaumont et al. “Vers l’automatisation de la gestion du cycle de vie des jumeaux numériques”. In: *Actes de la 44ème édition de la conférence INFORSID*. Toulouse, France, May 2026, pp. 55–57. URL: https://inforsid2026.sciencesconf.org/data/Actes_INFORSID2026.pdf.
- [C6] Gwendal Beaumont et al. “Towards Automating the Life Cycle Management of Digital Twins”. In: *Conceptual Modeling*. Ed. by Dominik Bork et al. Vol. 16189. ER 2025. Poitiers, France: Springer Nature Switzerland, Oct. 2025, pp. 412–430. ISBN: 978-3-032-08623-5. DOI: 10.1007/978-3-032-08623-5_22.
- [C7] Gwendal Beaumont et al. “Towards Re-Engineering Digital Twins: Preliminary Experiments on Three Use Cases”. In: *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion ’24. Linz, Austria: Association for Computing Machinery, Sept. 2024, pp. 453–458. ISBN: 979-8-4007-0622-6. DOI: 10.1145/3652620.3688259.
- [C8] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. “On the Necessity of Human Decision-Making Errors to Explain Vaccination Rates for Covid-19: An Agent-Based Modeling Study”. In: *2023 Annual Modeling and Simulation Conference (ANNSIM)*. New York, NY, USA: IEEE Computer Society, May 2023, pp. 413–424. ISBN: 978-1-7138-7328-0. URL: <https://ieeexplore.ieee.org/document/10155377>.